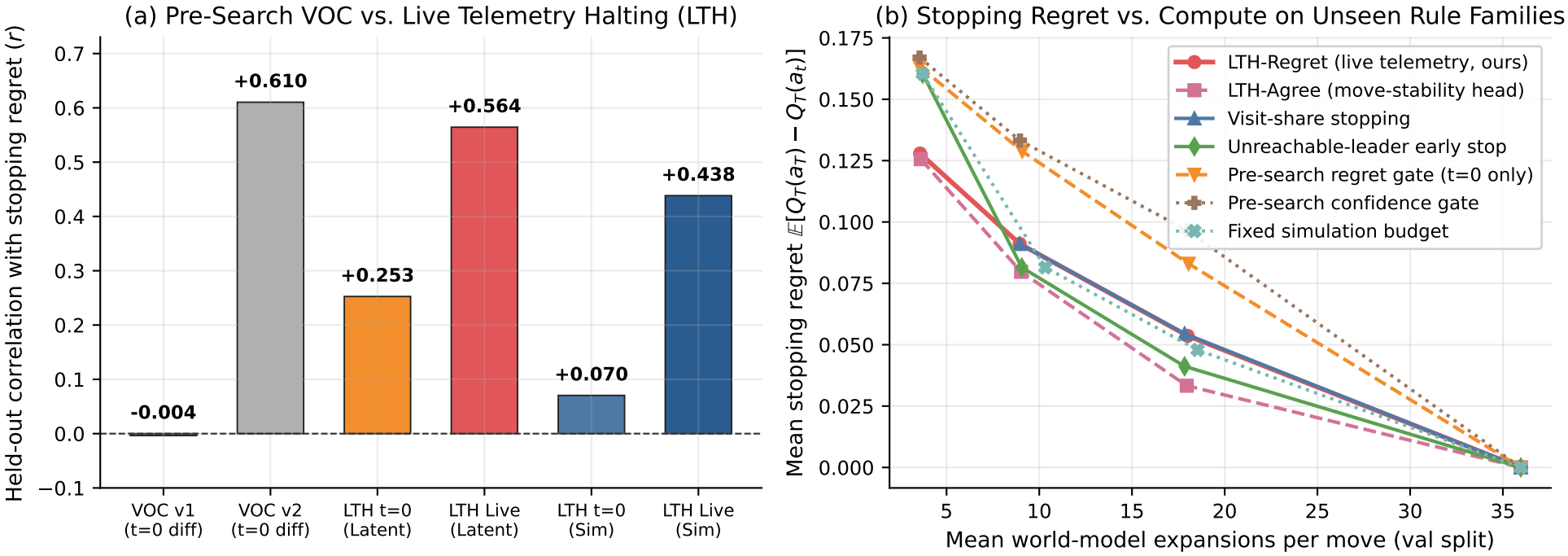
(a) Pre-Search VOC vs. Live Telemetry Halting (LTH)
Held-out correlation with stopping regret (r)
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
-0.004
+0.610
+0.253
+0.564
+0.070
+0.438
VOC v1 (t=0 diff)
VOC v2 (t=0 diff)
LTH t=0 (Latent)
LTH Live (Latent)
LTH t=0 (Sim)
LTH Live (Sim)
(b) Stopping Regret vs. Compute on Unseen Rule Families
Mean stopping regret $\mathbb{E}[Q_T(a_T) - Q_T(a_t)]$
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
5
10
15
20
25
30
35
Mean world-model expansions per move (val split)
LTH-Regret (live telemetry, ours)
LTH-Agree (move-stability head)
Visit-share stopping
Unreachable-leader early stop
Pre-search regret gate (t=0 only)
Pre-search confidence gate
Fixed simulation budget